

NETHRAJA KANDULA

FOOD SCIENTIST | FOOD SAFETY & APPLIED R&D

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PROFESSIONAL SUMMARY

Food scientist researcher with 3+ years of applied laboratory experience in food safety, microbiology, molecular assay development, shelf-life research, and data-driven process troubleshooting. Experienced in designing controlled experiments, optimizing methods, interpreting quantitative results, preparing SOP-style documentation, and managing multiple projects from concept through reporting. IHA-trained in HACCP principles and GMPs, with a strong interest in translating food science into practical, scalable solutions for meat and value-added food products.

CORE COMPETENCIES

- Experimental Design & Execution
- HACCP Principles & GMPs
- Food Microbiology & Shelf-Life
- Data Analysis & Reporting
- Product / Process Troubleshooting
- Scientific Team Communication

PROFESSIONAL EXPERIENCE

Jan 2023–July 2026

Graduate Research Assistant | Florida State University, Tallahassee, FL

- Develop and standardize digital PCR methods for detection and quantitative estimation of *Salmonella* in whole chicken carcass rinsate, chicken breast and ground turkey, including study design, matrix-specific optimization, controls, and performance evaluation.
- Characterize the growth behavior of multiple *Salmonella* strains and analyze quantitative datasets to support method development, decision-making, and reproducible experimental outcomes.
- Apply whole-genome sequencing, CRISPR-based genomic analysis, and hybrid-capture next-generation sequencing to support *Salmonella* characterization and serotyping in poultry products.
- Developed molecular assays for seafood species identification and supported external implementation through protocol demonstrations and technical training.
- Coordinate concurrent research activities, troubleshoot assay and workflow inconsistencies, maintain detailed experimental records, and communicate findings through manuscripts, presentations, and sponsor-facing updates.
- Served as Co-Investigator on an externally funded \$79,763 project focused on practical *Salmonella* detection and estimation methods for poultry samples.

Graduate Teaching Assistant / Instructor | Florida State University, Tallahassee, FL

Jan 2024–Dec 2025

- Taught the scientific basis of ingredient functionality, cooking processes, food quality, and safety through hands-on food laboratory instruction.
- Designed course materials and guided students in evaluating how formulation and processing variables influence product characteristics and final quality.
- Managed laboratory activities, documentation, safety expectations, and real-time troubleshooting in a fast-paced instructional environment.

Mar 2022–Jun 2022

Research Intern | Samath Global Food Consultants, Telangana, India

- Formulated seasoning blends for potato chip applications and developed flavor variants for protein powder products.
- Assisted with bench-scale formulation adjustments and sensory-focused evaluation of ingredient combinations to improve product acceptability.

SELECTED APPLIED RESEARCH & LEADERSHIP

- **Shelf-Life and Preservation:** Conducted research on shelf-life and microbial community dynamics of reduced-oxygen packaged seafood products; co-authored research on antioxidant and antibacterial active films for salmon preservation.
- **Method Transfer and Technical Support:** Coordinated and delivered on-site assay demonstrations at federal laboratories, explaining sample handling, controls, workflow execution, result interpretation, and troubleshooting.
- **Research Leadership:** Trained undergraduate researchers in aseptic technique, good laboratory practices, accurate documentation, safety, workflow execution, and independent troubleshooting.
- **Innovation and Commercialization:** Completed the NSF I-Corps Mid-South regional program and served as PI on a \$2,500 microgrant, gaining experience in customer discovery and translating technical research into practical value propositions.

EDUCATION

Ph.D., Food Science Florida State University, Tallahassee, FL Dissertation defended June 2026 GPA: 3.58/4.00	2022–2026
B.Tech., Food Technology Osmania University, Telangana, India CGPA: 8.2/10	2017–2021

TECHNICAL SKILLS

Food & Microbiology: Aseptic technique; microbial culture; selective plating; colony enumeration; growth-curve studies; poultry and seafood matrices; shelf-life studies

Molecular & Genomic: Digital PCR; real-time PCR; DNA extraction; assay optimization; next-generation sequencing; whole-genome sequencing; CRISPR analysis

R&D / Quality: Bench-scale experimental trials; formulation support; root-cause troubleshooting; technical documentation; SOP development; HACCP principles; GMPs

Data & Software: Python; GraphPad Prism; SPSS; MATLAB; Microsoft Excel and Office; CLC Genomics Workbench; DRAGEN

SELECTED PUBLICATIONS

- **Kandula, N.** et al. (2025). Rapid rhPCR lateral flow assay for detection of Pacific white shrimp. *Journal of Food Composition and Analysis*, 144, 107648.
- **Kandula, N.** et al. (2024). Varying amplification-efficiency real-time PCR assay for detection and estimation of *Salmonella* in poultry rinse. *Food Bioscience*, 61, 104777.
- Velez, F. J. *, **Kandula, N.*** et al. (2024). Digital PCR assay for detection and estimation of *Salmonella* contamination levels in poultry rinse. *Current Research in Food Science*, 9, 100807. (*Equal contribution)
- Cheng, J., **Kandula, N.** et al. (2024). Pea protein isolate-based active films for salmon preservation. *Current Research in Food Science*, 9, 100936.

CERTIFICATIONS & RECOGNITION

IHA-Accredited Certified HACCP Principles and GMP Training for Food Processors and Manufacturers (University of Georgia, 2025)
| NSF I-Corps Mid-South Regional Cohort (2024) | Recipient of three FSU graduate scholarships (2023–2025)